

SOFTWARE DESIGN

Collective 4 – SPRINT REVIEWS

SPRINT 1 REVIEW

Demonstration Summary

The following was demonstrated at the Sprint 1 review:

- Google OAuth login flow with role-based redirect to admin or member dashboard
- Stokvel group creation form submitting successfully to the database
- Member invite form sending invite links via email
- Contributions page displaying contribution history for a logged-in member
- Landing page and base navigation layout

Sprint Outcome

All four Sprint 1 user stories were completed and demonstrated successfully. The core authentication and group management foundation was in place by the review date. The team noted commitment challenges during the sprint and agreed to improve individual follow-through going into Sprint 2.

SPRINT 2 REVIEW

Demonstration Summary

The following was demonstrated at the Sprint 2 review:

- SA prime lending rate and repo rate pages fetching and displaying live SARB data
- Role assignment by admin via dropdown on the admin dashboard
- Token-based invite acceptance flow for both member and treasurer roles
- Treasurer dashboard with financial overview

- Logout functionality clearing session and redirecting to the login page
- UML Use Case Diagram and Class Diagram presented to the tutor

Sprint Outcome

All Sprint 2 user stories were completed and demonstrated. The UML diagrams were introduced this sprint and reviewed by the tutor. The team identified that code coverage needed improvement and committed to addressing this in Sprint 3.

SPRINT 3 REVIEW

Demonstration Summary

The following was demonstrated at the Sprint 3 review:

- Online contribution payment via PayFast sandbox, with status update on return
- Treasurer confirming and flagging contributions on the contributions dashboard
- Meeting scheduling form, minutes recording, and meeting deletion
- Payout schedule with drag-to-order and position display for members
- Analytics dashboard with three report types and CSV export
- GitHub Actions CI/CD pipeline with automated Jest test runs and CodeCov integration
- Updated UML diagrams including new sequence diagrams for contributions, payments, meetings, and payouts

Sprint Outcome

All 11 Sprint 3 user stories were completed and demonstrated. The PayFast integration, analytics module, and CI/CD pipeline were the most significant additions this sprint. Code coverage was being monitored and the team committed to reaching at least 80% by Sprint 4.

SPRINT 4 REVIEW

Demonstration Summary

The following was demonstrated at the Sprint 4 review:

- Payout disbursement button on the treasurer payouts dashboard, recording payout receipt per member
- Manual contribution recording form for offline and cash payments
- PDF export of the analytics dashboard
- Profile page with name editing and read-only display of email, group, and role
- Notifications page aggregating group activity into a scrollable list
- Final UML diagrams: updated Component Diagram, Deployment Diagram, and all Sequence Diagrams covering Sprint 4 features
- Code coverage improvement from 67% toward the 80% target via additional Jest test files

Sprint Outcome

All 5 Sprint 4 user stories were completed and demonstrated. The platform was fully deployed on Azure and all features across all four sprints were functional on the live site. The final UML diagrams were approved by the team and committed to the repository. The project was prepared for final submission on Moodle by 22 May 2026.

2026.